MIL-P-36855A
27 July 1987
SUPERSEDING
MIL-P-36855
14 June 1972

MILITARY SPECIFICATION

PLIERS, DENTAL, (Amalgam Reinforcement Pin)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers wire cutting pliers with a plastic insert in the jaws for use in amalgam pin reinforcement technique.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	Boxes, Fiberboard.

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RSTS, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Federal

FED-STD-66	Steel, Chemical Composition and Hardenability.
FED-STD-151	Metal, Test Methods.
FED-STD-376	Preferred Metric units for general use by the Federal Government.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads.
MIL-STD-794	Parts and Equipment, Procedures for Packaging of.

DRAWINGS

Defense Personnel Support Center (DPSC)

24117	Pliers, Dental, Amalgam, Reinforcement Pin, 4-7/8 In. (With Plastic Insert In Each Jaw)

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS are the issues of the non-Government documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 18	Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
E 92	Standard Test Method for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa 19103).

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Drawings. The drawings forming a part of this specification are specification control drawings. The supplier is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in improper design, function, or operation, the supplier shall provide tolerances within those prescribed on the drawings to insure proper design function, and operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval from the contracting officer.

3.2 Materials.

3.2.1 Pliers. Pliers shall be fabricated of chrome-vanadium steel conforming to the chemical requirements for AISI 6150 steel of FED-STD-66, or carbon steel conforming to the chemical requirements for SAE 1055 steel of FED-STD-66.

3.2.2 Plastic inserts for jaws. The insert shall be a resilient, vinyl plastisol.

3.2.3 Plastic handle covering. The handle covering shall be a colored polyvinyl chloride type of plastic.

3.3 Hardness.

3.3.1 Steel. After proper stress relieving, the Rockwell hardness shall be C36 to 50 for the body, and C55 to 65 for the cutting edge of the jaws, when tested in accordance with 4.4.2.

3.3.2 Plastisol. Vinyl plastisol (end item, with insert applied to pliers) shall have a hardness equal to a 10 second reading of 50.0 to 75.0 points when tested in accordance with 4.4.3.1.

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3.3.3. Plastisol after aging. After aging, the vinyl plastisol hardness (end item, with insert applied to pliers) shall increase no more than 10.0 points when tested in accordance with 4.4.3.2.

3.4 Distortion. The handles of the pliers shall show no pliability or distortion under normal usage.

3.5 Style, design and dimensions. The pliers shall be a diagonal cutting type. Pliers shall be specifically designed to be suitable for use in the dental operatory during the treatment of patients, in support of the Markley pin-retention technique. Pliers shall be in accordance with Defense Personnel Support Center (DPSC) Drawing 24117. The pliers shall have a plastic insert in each jaw which shall serve as a cushion. The pliers shall be capable of cutting, gripping, and holding one cut end of 0.022 and 0.025 inch diameter threaded wire (NSN 6520-00-935-1096 and 6520-00-935-1095).

3.6 Construction.

3.6.1 Jaws. The inner jaws of the pliers shall be serrated and the insert (3.2.2) shall be primed for good adhesion to steel and affixed to each inner jaw.

3.6.2 Locks. The joint of the pliers shall be the lap joint type. Locks shall be accurately fitted, without stiffness, and without crevices, burrs, or sharp edges (outer edges of joints shall be properly beveled). The edges of the joints shall be properly beveled to prevent pinching of tissue. Pins shall be permanently secured.

3.6.3 Handles. The handles shall be tightly encased in a neat, resilient, smooth-surfaced, plastic (3.2.3) which will provide a comfortable, non-skid grip.

3.6.4 Cutting edges. Cutting edges shall be properly sharpened to the required type of edge and shall be free from burrs or feathered edges. The pliers shall function correctly for the particular work for which they are intended. Pliers shall be capable of cutting 0.022 and 0.025 inch diameter threaded, stainless steel wire (see 3.5) without causing an increase in the diameter of the wire at the cut ends when tested as specified in accordance with 4.4.4.

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3.7 Finish. Finish on all edges and surfaces shall be uniform, and free of burrs, sharp edges (except where required), crevices, grind marks, rough areas, cracks, pits, voids and overlaps. Exposed portions of handles shall be finely ground and polished. The diagonal cutting edges shall be finely honed. The beaks and head shall be polished to a bright or mirror finish. As an alternate, the pliers shall be bright chrome plated.

3.8 Identification marking. Marking shall be on the (outside, front or back face) metal portion (joint area). The marking shall be permanent and legible (chemical or electro-chemical etching shall be considered acceptable). Pliers made in U.S.A. shall be marked with the name or registered trademark of the contractor. Pliers of foreign (other than U.S.) manufacture shall be marked with the name of the country of origin, together with the name or registered trademark of the contractor.

3.9 Workmanship. Instruments shall be free from defects which detract from their appearance or impair their serviceability, proper functioning, and intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.4 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in Tables I and III (see 4.2.1 and 4.3) and Table II (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination for the pliers shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in Table I. The unit of product for examination purposes shall be one pliers.

TABLE I. Inspection and acceptable quality levels

Type defect	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-1	1.5
Minor	II	2.5

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4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II. The acceptance number shall be zero for all sample sizes. The unit of product for test purposes shall be one pliers. Where feasible, the same sample unit may be used for the determination of two or more test characteristics.

Table II. Testing Criteria

Characteristic	Reqmt	Test Procedure	Unit of Product	Inspection Level
Composition (Steel)	3.2.1	4.4.1	1 pliers	S-2
Hardness (Steel)	3.3.1	4.4.2	1 pliers	S-2
Hardness (Plastisol)	3.3.2	4.4.3.1	1 pliers	S-2*
Hardness, Simulating aging (Plastisol)	3.3.3	4.4.3.2	1 pliers	S-2*
Cutting Steel Wire	3.6.4	4.4.5	1 pliers	S-2

*These tests shall be done in sequence (see 4.4.3.1 and 4.4.3.2) on the same sample unit.

4.2.3 Raw materials. Sufficient quantity of each lot of metal used in the construction of the pliers shall be selected for chemical analysis.

4.3 Examination. The pliers shall be examined for defects including, but not limited to, those listed in Table III.

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TABLE III. Classification of defects

Category	Defects
Major A	
101	Style and design not as specified.
102	Jaw plastic insert inadequately secured, loose or missing.
103	Handles not tightly encased in plastic.
104	Handle plastic not resilient and smooth surfaced.
105	Metal surface finish not as specified.
106	Edges and surfaces not free from burrs, crevices, pits, voids and overlaps.
107	Plastic parts not free from patches, cracks, holes, voids, or other imperfections.
108	Joint of pliers not free from stiffness.
109	Joint of pliers has excessive play or is excessively loose.
110	Edges of the joint of pliers sharp (not properly beveled).
111	Workmanship not as specified.
112	Cutting edges not sharp.
113	Cutting edges not properly aligned.
114	Members of pliers not permanently secured.
Major B	
151	Dimensions not as specified.
Minor	
201	Pliers not permanently and legibly marked with the name or trademark of the contractor.
202	Marking not properly located.
203	Metal surfaces not free from grind marks and rough areas.
204	Plating not free of peeling or blisters.

4.4 Tests.

4.4.1 Chemical analysis. In the absence of Certificates of Quality, chemical composition of the steel used in the manufacture of the pliers shall be determined in accordance with method 111 or 112 or FED-STD-151. In the event of conflict, method 111 shall govern.

4.4.2 Hardness (Steel). Hardness shall be determined in accordance with ASTM Method E 18 (Rockwell) or ASTM Method E 92 (Vickers).

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4.4.3 Plastisol tests. The following plastisol tests shall be performed after the plastisol inserts or cushions have been applied to the throat of the jaws.

4.4.3.1 Hardness test. The hardness characteristic of the vinyl plastisol shall be tested using a Shore Type A-2 Durometer, or equivalent testing equipment. Firm pressure shall be used to seat the instrument against the specimen. Penetration shall be maintained and the reading noted at 10 seconds. Three readings shall be averaged for each unit of product tested. Defective readings shall be discarded and additional readings taken. A new spot shall be penetrated for each reading.

4.4.3.2 Simulated aging test. After being oven aged at 158.0 ± 4.0 degrees Fahrenheit (^{o}F.) for 48 hours, the sample shall be removed from the oven and allowed to cool to room temperature. The hardness test of 4.4.3.1. shall again be performed.

4.4.4 Cutting test. The pliers shall cut the 0.022 and 0.025 inch diameter stainless steel threaded wire as specified (see 3.6.4). Three separate consecutive tests shall be performed for each pliers. Each test shall consist of making one cut through the test material in a different portion of the jaws.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Pliers shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. Each pair of pliers shall be packaged in accordance with MIL-P-116, method IC-1 or IC-3.

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5.1.1.4 Intermediate package. Twelve unit packages shall be packaged in box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C.

5.1.2.1 Unit package. Preservation for pliers shall conform to the requirements of MIL-STD-794 for level C.

5.1.2.2 Intermediate package. Twelve unit packages shall be packaged in accordance with the requirements of MIL-STD-794 for level C.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Twelve intermediate packages (144 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2, PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Twelve intermediate packages (144 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1, or PPP-B-636, class domestic, closure shall be as specified in the applicable box specification.

5.2.3 Level C. Twelve intermediate packages (144 units) shall be packed in accordance with the requirements of MIL-STD-794 for level C.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

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5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination exceed 250 pounds or 20 cubic feet. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 20 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 50 inches in height, 48 inches in width, and 40 inches in length. Pallet shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Pliers packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. Pliers packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4. General

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 National Stock Number (NSN) coverage. This specification covers the following item listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6520-00-935-1094	PLIERS, DENTAL, Amalgam Reinforcement Pin, 4-7/8 inch: With plastic insert in each jaw.

6.3 Subject term (key word) listing:
Amalgam Reinforcement Pin
Dental Pliers

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Review Activities:

Army - MD
Navy - MS
Air Force - 03

Civil Agency Coordinating Activities:
VA-OSS
PHS
FDA-MPQAS

48636

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

Project No. 6520-1932

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-36855A	PLIERS, DENTAL, AMALGAM REINFORCEMENT PIN

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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